

AUDITRACK: COMPLIANCE AND AUDIT SYSTEM

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CHAPTER I

INTRODUCTION

The Augustinian Sisters of Our Lady of Consolation (ASOLC) institutions are committed to maintaining high standards of academic integrity, institutional accountability, and regulatory compliance across their member schools. To ensure adherence to institutional policies and government regulations, internal compliance audits are regularly conducted within academic and administrative departments. These audits serve as essential mechanisms for evaluating compliance status, identifying operational gaps, and promoting continuous improvement. However, many ASOLC institutions continue to rely on manual, paper-based, and spreadsheet-supported audit processes, which often result in fragmented documentation, delayed reporting, and inconsistencies in record management.

The current approach to compliance auditing involves manual data encoding, physical document storage, and decentralized file management. These practices increase the risk of data loss, duplication, and human error while limiting real-time visibility of audit progress. The absence of a centralized digital platform makes it difficult for authorized personnel to efficiently retrieve audit records, monitor compliance status, and generate timely reports. As institutional operations grow more complex, the need for a structured, secure, and technology-driven audit management system becomes increasingly evident.

In response to these challenges, this study proposes the design and development of a centralized and automated Compliance Audit System for ASOLC institutions. The proposed system is a web-based application that digitizes audit forms, automates workflow processes, and stores audit records in a centralized database. It incorporates role-based access control to ensure that administrators, internal auditors, and authorized personnel can securely access and manage audit information according to their assigned privileges. Additionally, the system provides real-time monitoring features, including dashboards and status indicators, to improve transparency and support timely decision-making.

By replacing manual and paper-based procedures with an integrated digital platform, the Compliance Audit System aims to enhance efficiency, improve data accuracy, strengthen data integrity, and streamline compliance documentation across departments. Ultimately, this initiative supports ASOLC's commitment to institutional governance, accountability, and continuous quality improvement through structured and technology-enabled compliance management

PROJECT DESCRIPTION

The Compliance Audit System is a web-based application designed to modernize and automate institutional compliance audit processes. The system replaces existing manual, paper-based, and spreadsheet-supported audit procedures with a centralized digital platform that improves efficiency, data accuracy, accessibility, and transparency in managing compliance-related activities.

The system provides a structured environment where audit records are stored in a centralized database, allowing authorized users to create, review, manage, and retrieve audit documentation securely. Through digitization of audit forms and documentation, the system minimizes reliance on physical paperwork and reduces the risks of data loss, duplication, and misfiling. Role-based access control ensures that administrators, internal auditors, compliance officers, and authorized department personnel can access information according to their assigned privileges while maintaining data integrity and confidentiality.

Key features of the system include automated audit workflow management, digital data collection, evidence attachment and storage, real-time monitoring of audit progress through dashboards and status indicators, and automated report generation. These functions streamline audit procedures, reduce processing delays, and minimize human error commonly associated with manual systems. By centralizing compliance data and automating routine processes, the system enhances operational efficiency and supports consistent documentation practices across departments. The Compliance Audit System also serves as a decision-support tool for institutional governance. By providing up-to-date audit information and real-time visibility of compliance status,

the system enables administrators and authorized personnel to identify gaps promptly, implement corrective actions, and strengthen institutional accountability.

Overall, the system contributes to the institution's digital transformation efforts by delivering a more organized, reliable, and transparent approach to compliance audit management. It supports internal compliance monitoring within the institution's existing technological infrastructure and is designed to improve the overall effectiveness of audit operations while maintaining data security and structured information management.

PROJECT CONTEXT

The rapid advancement of information technology has encouraged educational institutions to adopt digital solutions that improve operational efficiency and information management. One important area that benefits from digital transformation is compliance auditing, which ensures that institutional policies, standards, and regulatory requirements are consistently followed. However, many institutions still rely on manual, paper-based documentation and spreadsheet applications to manage audit records, resulting in inefficient workflows, difficulty in tracking audit progress, data inconsistencies, and delays in generating reports.

To address these challenges, the Compliance Audit System was developed as a web-based application that automates and centralizes compliance audit processes. The system provides a secure platform where authorized users can create, manage, review, and retrieve audit records through a centralized database. It also incorporates role-based access control, digital documentation, evidence storage, real-time monitoring through dashboards, and automated report generation. These features help reduce manual work, minimize human error, improve data accuracy, and ensure that audit information is organized and easily accessible.

In addition, the system supports administrators, internal auditors, compliance officers, and authorized department personnel in monitoring compliance activities more efficiently. By providing real-time access to audit status and compliance information, the system enables timely identification of compliance gaps and facilitates informed decision-making. Overall, the Compliance Audit System contributes to the institution's digital transformation by promoting a more efficient,

transparent, secure, and reliable approach to compliance audit management while improving the effectiveness of institutional governance.

OBJECTIVES OF THE STUDY

The primary objective of this study is to design and develop a centralized and automated Compliance Audit System that addresses the inefficiencies and challenges associated with existing manual and paper-based audit processes. Specifically, the study aims to:

- Design and implement a unified digital platform that centralizes the storage, management, and retrieval of compliance audit records to ensure easier access, improved organization, and reduced risk of data loss or misfiling.
- Automate audit workflows and documentation procedures to minimize processing delays, reduce human error, and enhance the accuracy and completeness of audit data across academic and administrative units.
- Enable real-time monitoring of compliance activities through dashboards, automated status updates, and notification mechanisms to improve transparency and allow timely identification and resolution of compliance issues.
- Implement role-based access control and data security measures to ensure that audit information is accessible only to authorized users while maintaining data integrity and confidentiality.

Through the development of this system, the study seeks to modernize institutional compliance audit operations, improve reporting reliability, and promote a more efficient, transparent, and accountable compliance management process.

SCOPE AND LIMITATIONS OF THE STUDY

This study focuses on the design and development of a centralized and automated Compliance Audit System for institutional use. The system aims to digitize and streamline compliance audit processes by providing a structured platform for managing audit documentation, tracking audit activities, and monitoring compliance status across departments. It seeks to enhance efficiency, data accuracy, accessibility, and organization of audit-related records.

- The study covers system requirement analysis, design, and development of a web-based application featuring digital audit forms, automated workflow tracking, real-time monitoring, role-based access control, and centralized storage of audit reports and supporting documents. The system will be tested within selected departments involved in internal compliance activities.
- The system supports key user roles, including administrators, internal auditors, compliance officers, and authorized department personnel responsible for submitting, reviewing, and managing audit data.

However, the study is limited to internal compliance audit operations and does not include external audits conducted by accrediting agencies or government institutions. Financial auditing processes, advanced risk assessment requiring specialized accounting standards, and integration with enterprise resource planning (ERP) systems are excluded from the current scope.

- The system will be deployed using cloud-based hosting and database services, such as web hosting platforms and online database providers, accessible through an internet connection. The study includes the development of both web-based and mobile application platforms to improve accessibility and usability for authorized users. Furthermore, the system's accuracy and effectiveness depend on the correctness and completeness of data entered by authorized users.

DEFINITION OF TERMS

For better understanding throughout the study, the terms are defined as follows:

Accessibility

Conceptually, means how easy it is for authorized users to access and use information in a system.

Operationally, it is the system's ability to allow internal auditors, administrators, and department heads to log in successfully, retrieve audit records quickly, and view real-time data based on their assigned access rights.

Administrator

Conceptually, a person who manages and controls a system.

Operationally, the designated user who manages roles, system settings, audit approvals, and ensures data security in the Compliance Audit System.

Audit Records

Conceptually, documents and reports used to check if an organization follows rules and policies.

Operationally, the digital files, electronic forms, and stored audit findings saved in the system's database that can be searched and filtered.

Compliance Audit System

Conceptually, a digital platform used to organize and monitor compliance auditing activities.

Operationally, the web-based system developed to manage audit documents, automate workflows, monitor audit progress, and generate reports.

Database

Conceptually, a structured place where digital information is stored.

Operationally, the system's MySQL relational database where audit data, user details, and compliance records are stored and managed using SQL.

Digitization

Conceptually, converting paper-based processes into electronic form.

Operationally, the system's use of electronic audit forms, digital uploads, and automated tracking to reduce physical paperwork.

Internal Auditor

Conceptually, a person who examines records to ensure rules and standards are followed.

Operationally, a system user who creates audits, reviews submissions, records findings, and generates compliance reports.

Mobile-Based Application

Conceptually, a software application designed to run on smartphones or tablets for convenient access anywhere.

Operationally, the mobile version of the Compliance Audit System that allows authorized users to view audit records, receive notifications, and update compliance data using their mobile devices.

Paper-Based System

Conceptually, a traditional system that uses printed documents and manual filing.

Operationally, the old method of using printed audit checklists, physical folders, and manual tracking before the implementation of the Compliance Audit System.

Real-Time Monitoring

Conceptually, tracking activities as they happen.

Operationally, the system feature that updates dashboards, status indicators, and notifications immediately when audit data is entered or modified.

User Interface

Conceptually, the visual part of a system that allows users to interact with it.

Operationally, the web-based forms, dashboards, menus, and buttons of the Compliance Audit System evaluated based on usability and responsiveness.

Web-Based Application

Conceptually, a program that runs through an internet browser without needing installation.

Operationally, the Compliance Audit System accessed through a secure URL using standard web browsers, operating over HTTPS and hosted on institutional servers.

CHAPTER II

TECHNICAL BACKGROUND

This chapter presents the requirements necessary to build and run the proposed Compliance Audit System. It outlines the essential hardware, software, and peopleware components that collectively enable the system to function efficiently. Furthermore, it discusses the testing procedures that will be conducted to ensure the system's reliability, functionality, and compatibility across different devices, including computers and mobile platforms.

Software Specification

The software applications that are going to be used in the development of the system are the following

Software	Details	Description
Microsoft Edge	Version 146.0.3856.72	Microsoft Edge is a web browser developed by Microsoft that is used to access and browse websites on the internet. It comes pre-installed on Windows devices.

XAMPP	• Version 8.2.12	XAMPP is a free tool that allows developers to create and test websites on their own computer. It includes essential components like a web server and database in one package.
Visual Studio Code	• Version 1.112	Visual Studio Code is a free code editor used for writing and editing programs. It supports many programming languages and includes helpful features for development.

Table 2.1. Software Specification

Hardware Specification

The hardware specifications required for the development of the system are the following:

Hardware	Details	Description
CPU	Intel Core i3-1115G4 processor Dual-core 3 GHz	The Central Processing Unit (CPU) is the main part of the computer that processes instructions and runs programs. It handles most of the system's operations.
RAM	16384MB DDR5	RAM (Random Access Memory) is the computer's short-term memory used to run applications and manage tasks while the system is in use. More RAM allows smoother multitasking.

Storage (HDD/SSD)	• 500GB NVMe M.2	Storage is where all files, applications, and the operating system are saved. NVMe SSDs provide fast data access and quicker system performance.
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Table 2.2. Hardware Specification

Peopleware Specification

Peopleware refers to the individuals involved in the development, implementation, and operation of the system. It emphasizes the role of users in ensuring that the system functions effectively through proper interaction, coordination, and task execution. In the proposed Compliance Audit System, different users are assigned specific roles and responsibilities to maintain efficiency, security, and accountability.

Peopleware	Description
System Administrator	Manages the overall system, including user accounts, system configurations, access control, and maintenance of the platform.

ASOLC Administrator	Oversees compliance reports and analytics across all institutions and supports decision-making at the organizational level.
Institution Administrator	Manages audit activities, users, and compliance records within a specific institution.
Internal Auditors	Conduct audits, evaluate compliance, input audit data, and generate audit reports based on institutional standards.
Department Personnel	Submit required documents, provide necessary information, and respond to audit findings within their respective departments.

Table 2.3. Peopleware Specification

Testing Plan

To ensure the reliability and effectiveness of the Compliance Audit System for ASOLC institutions, the system will undergo functionality, acceptance, and performance testing. Functionality testing will verify that features such as audit management, role-based access, evidence storage, dashboards, and report generation work correctly. Acceptance testing will evaluate whether administrators, auditors,

compliance officers, and authorized personnel find the system easy to use and suitable for compliance audit processes. Performance testing will assess the system's speed, stability, and responsiveness when handling large volumes of audit records. These testing procedures aim to identify and resolve issues to ensure the system operates efficiently, securely, and reliably.

Functionality Testing

Functional testing is conducted to ensure that all features of the Compliance Audit System operate according to the specified requirements. This includes verifying core functions such as user authentication, role-based access control, audit record management, evidence attachment and storage, compliance status updates, dashboard monitoring, and automated report generation. The objective is to confirm that each component performs accurately, consistently, securely, and without errors to support efficient and reliable compliance audit management.

Functionality Features

Functions	Features	Descriptions
1	Data Encoding	Allows authorized users to input, update, and manage compliance and audit-related information within the system.
2	Central Database	Stores all compliance documents, audit records, and user information in a

		centralized and organized database for easy access and management.
3	Role-Based Access Control	Restricts system access and features based on user roles such as administrator, staff, or office personnel.
4	Document Management	Enables uploading, viewing, organizing, and tracking of compliance and accreditation documents.
5	Audit Tracking	Monitors the status and progress of audit and compliance requirements across departments.
6	Search and Filter Functionality	Allows users to quickly locate documents and records using keywords, categories, or filters.
7	Notification System	Sends reminders or alerts for pending requirements, deadlines, or updates.
8	Report Generation	Generates reports related to compliance status, audit progress, and document submissions.

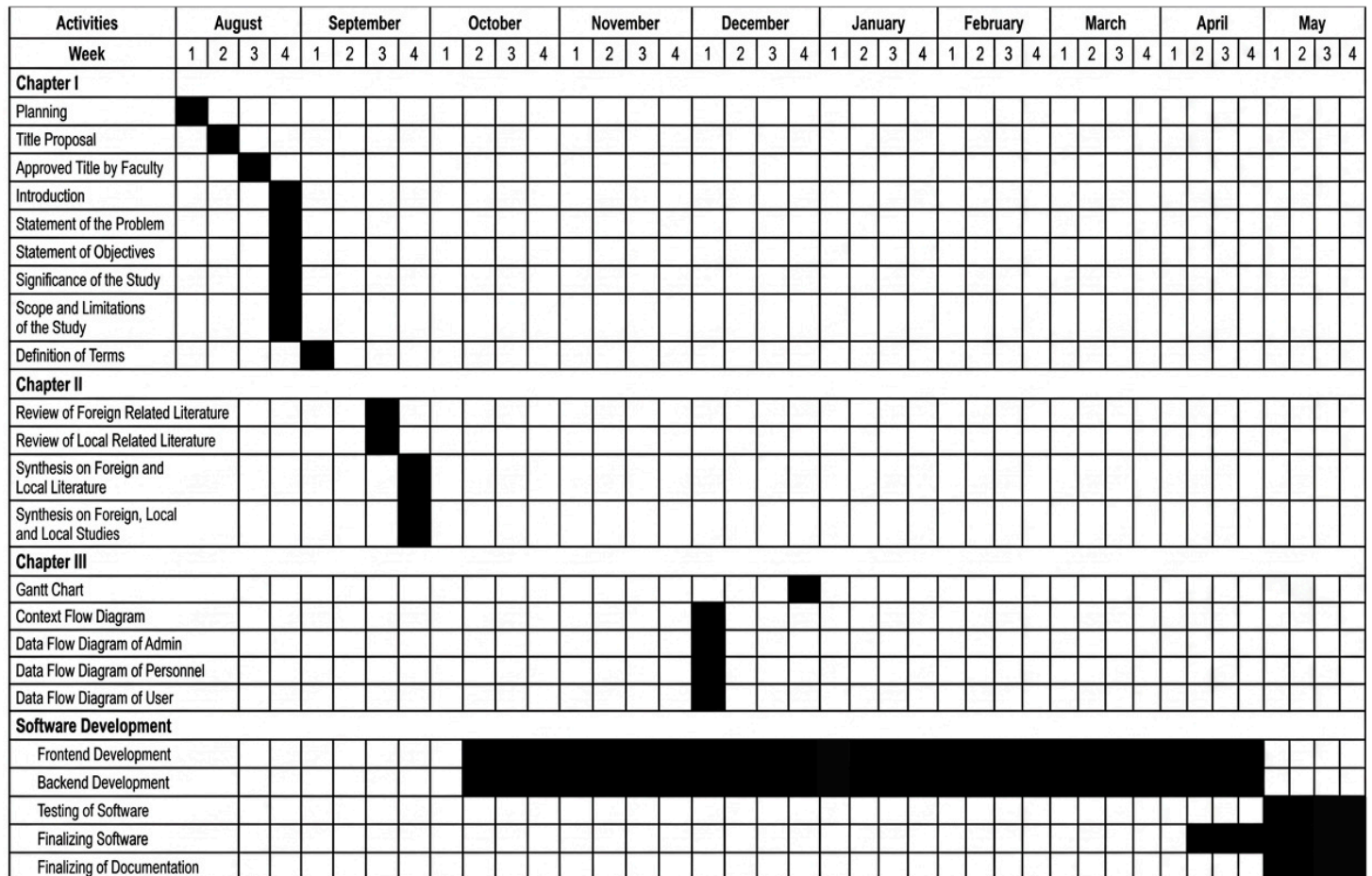
Acceptance Testing

Acceptance testing is conducted to determine whether the Compliance Audit System meets the needs of its intended users, specifically the administrators, internal auditors, compliance officers, and authorized personnel of ASOLC institutions. This testing phase focuses on evaluating the system's usability, functionality, reliability, and overall user satisfaction.

During this process, selected users will perform actual tasks such as managing audit records, uploading evidence, updating compliance status, monitoring audit progress through dashboards, and generating reports. They will assess whether the system is easy to use, understandable, secure, and efficient in supporting institutional compliance audit processes.

CHAPTER III

GANTT CHART



Environment

Locale

- **Location:** La Consolacion College Bacolod (LCCB), Educational institution environment, evaluated and implemented across selected academic and administrative departments actively conducting institutional compliance and audit operations.

Population of the Study

- **Administrators:** System Administrators and Institutional Executive Leadership responsible for system configuration, user role management, oversight, and policy-level decision-making.
- **Auditors & Compliance Personnel:** Lead Auditors, Internal Auditors, and Compliance Officers responsible for initiating audit schedules, conducting evaluations, attaching evidence, and monitoring corrective action plans.
- **Departmental Staff:** Department Heads, Administrative Officers, and designated staff members responsible for submitting digital audit documentation, responding to audit findings, and updating status reports.

Organizational Profile

- **Institution:** La Consolacion College Bacolod (LCCB), An educational institution transitioning from manual, paper-based, and spreadsheet-reliant audit workflows to a secure, web-based digital compliance audit platform.

System Roles Structure

- **System Administrator:** Full access to manage system settings, global access permissions, database configurations, and overall security protocols.
- **Compliance Administrator / Lead Auditor:** Manages audit creation, workflows, report generation, dashboard oversight, and user account management across departments.
- **Internal / Departmental Auditor:** Executes assigned compliance audits, conducts digital data collection, reviews evidence submissions, and updates audit progress.
- **Department Personnel / Staff:** Operational users authorized to upload evidence, view findings specific to their department, submit responses, and track corrective action statuses.

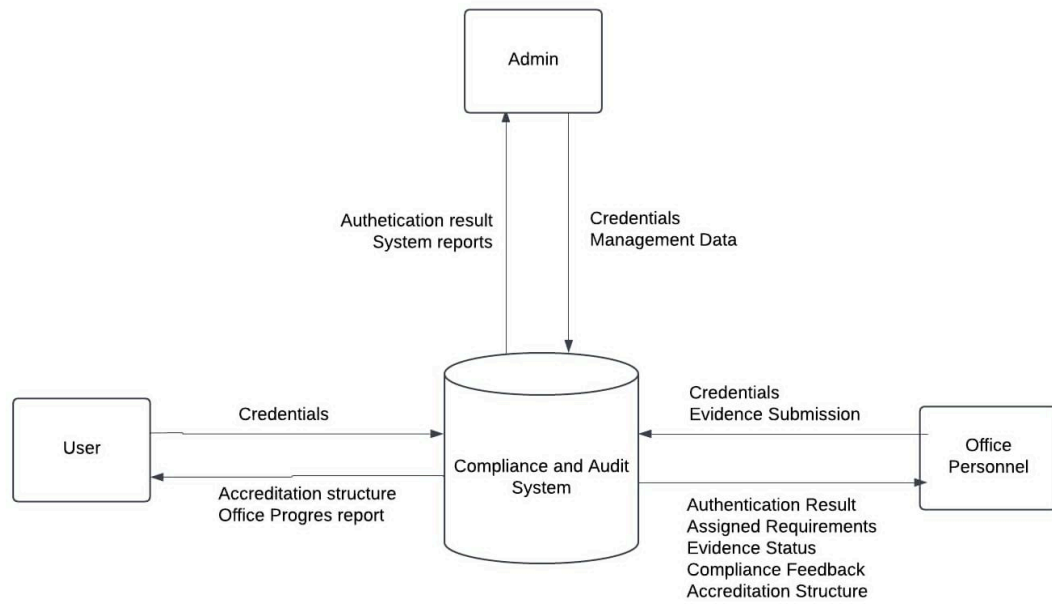
Requirement Specifications

Schedule Feasibility

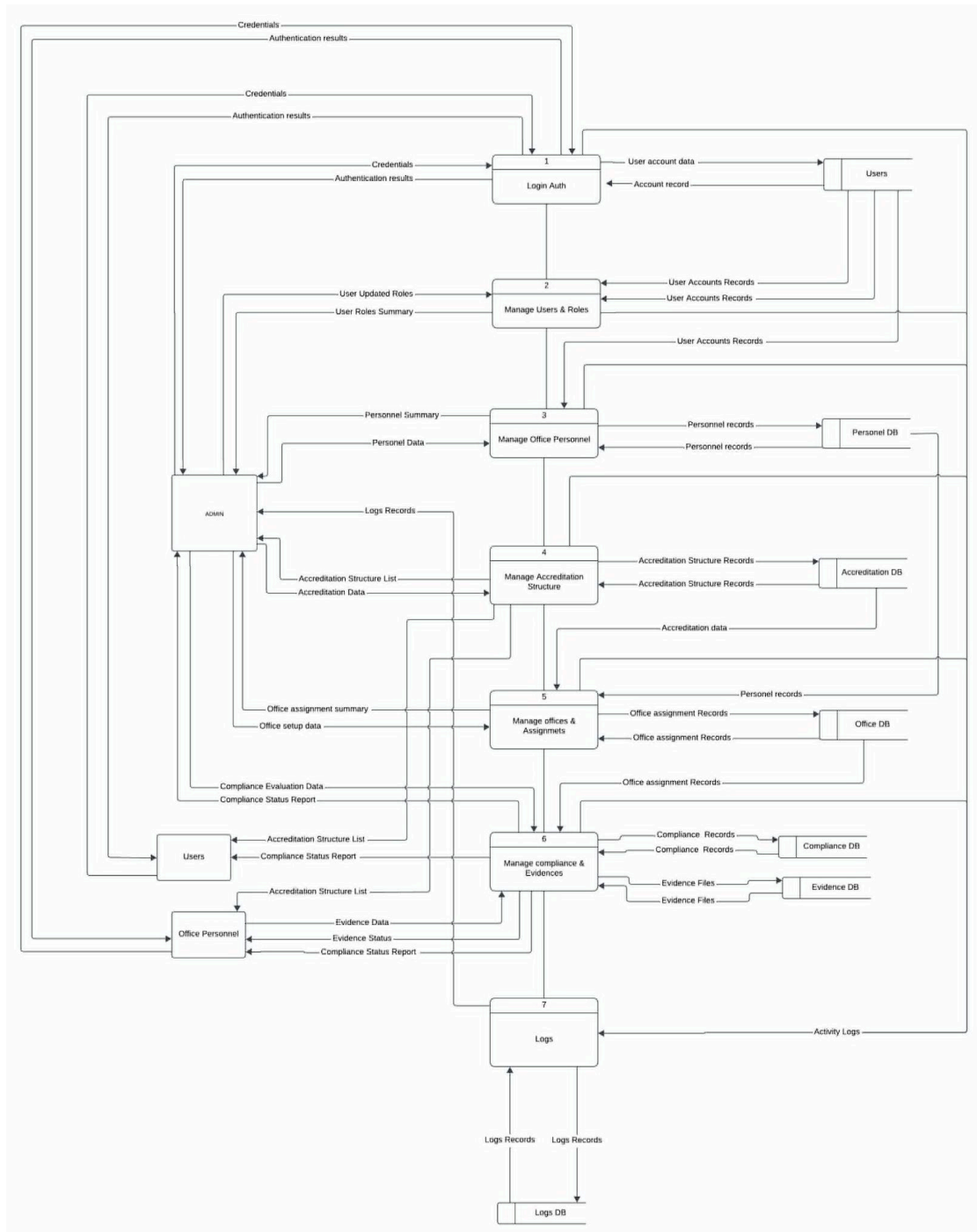
Project Phase	Tasks / Deliverables	Timeline
Planning & Documentation	Title Proposal, Approval, Chapter I (Introduction, Project Description, Project Context, Scope & Limitations)	Aug 2025 – Oct 2025
Technical Design	Chapter II (System Architecture, Requirements Specs, Testing Plan) & Chapter III (Data Flow Diagrams, Entity Relationship Diagram, Gantt Chart)	Nov 2025 – Feb 2026
Development	Web Application Development: Centralized Database Integration, Role-Based Access Control, Digital Form Digitization, Dashboard & Reporting Modules	Jan 2026 – Apr 2026
Testing & Deployment	Functional Testing, User Acceptance Testing (UAT with Auditors & Personnel), Data Security Verification, & System Deployment	Mar 2026 – Apr 2026
Finalization	Audit System Optimization, Final Revisions, & Manuscript Completion	Apr 2026 – May 2026

Requirements Modeling

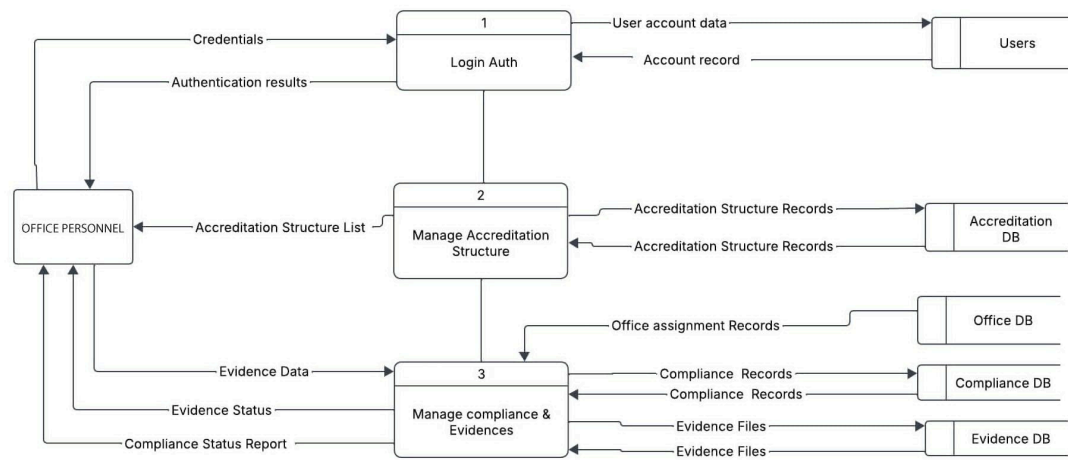
Context Flow Diagram



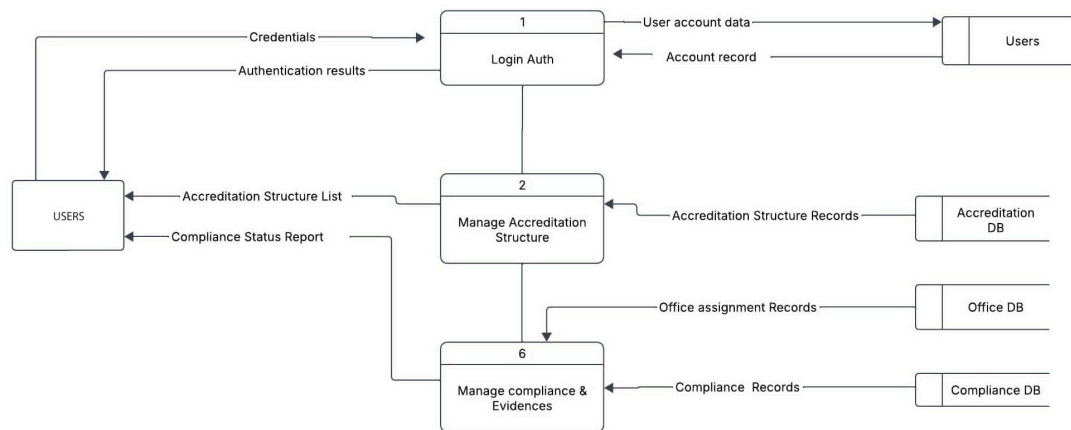
Data Flow Diagram of Admin



Data Flow Diagram of Personnel



Data Flow Diagram of User



Conceptual Framework

